

⑩ 13.1.5/ $0 = \frac{\partial}{\partial t} \langle \vec{R} \cdot \vec{P} \rangle = \langle [\vec{R} \cdot \vec{P}, \vec{H}] \rangle$

$$= \langle [xP_x + yP_y + zP_z, T + V] \rangle$$

Here $\langle [xP_x, \frac{P_x^2}{2m}] \rangle = \frac{1}{2m} \langle [x, P_x^2] P_x \rangle$

$$= i\hbar \langle \frac{P_x^2}{m} \rangle = 2i\hbar \langle T_x \rangle$$

$$\Rightarrow \langle [\vec{R} \cdot \vec{P}, T] \rangle = 2i\hbar \langle T \rangle$$

For the potential part, we work in the coordinate representation:

$$[xP_x, -\frac{e^2}{r}] \psi(x) = -e^2 \left(x \left(-i\hbar \frac{\partial}{\partial x} \right) \frac{1}{r} - \frac{x}{r} \left(-i\hbar \frac{\partial^2}{\partial x^2} \right) \right)$$

$$= i\hbar e^2 x \left(4 \frac{\partial}{\partial x} \frac{1}{r} \right)$$

$$= -i\hbar e^2 x 4 \frac{x}{r^3} = -i\hbar \frac{x^2}{r^2} \left(\frac{e^2}{r} 4 \right)$$

$$\Rightarrow [\vec{R} \cdot \vec{P}, V] = -i\hbar \frac{e^2}{r} = i\hbar V(x)$$

$$\Rightarrow \langle [\vec{R} \cdot \vec{P}, H] \rangle = (2\langle T \rangle + \langle V \rangle) i\hbar = 0$$

$$\Rightarrow \langle T \rangle = -\frac{1}{2} \langle V \rangle$$

⑤ 13.3.1/ The range corresponds to Compton wavelength

$$\lambda_C = \frac{h}{m_e c} \Rightarrow m_e c^2 = \frac{hc}{\lambda_C} = \frac{10^{-34} \times 3 \times 10^8}{10^{-15} \times 1.6 \times 10^{19}} \text{ eV}$$
$$\cong 200 \text{ MeV}$$

⑤ 13.3.3/ $-(E_2 - E_1) = -13.6 \left(1 - \frac{1}{4}\right) \text{ eV} = -10.2 \text{ eV}$

For $T = 300 \text{ K}$, $kT = \frac{1}{40} \text{ eV}$

$$\Rightarrow T = 6000 \text{ K}, \quad kT = \frac{1}{40} \times 20 \text{ eV} = \frac{1}{2} \text{ eV}$$

$$\Rightarrow \frac{P(n=2)}{P(n=1)} \cong 4 e^{-20} \rightarrow 0$$

$$T \sim 10^5 \text{ K}, \quad kT \cong \frac{1}{40} \times 400 \text{ eV} = 10 \text{ eV} \quad (T = 1.2 \times 10^5 \text{ K})$$

Then kT is comparable to 10.2 eV , and

$\frac{P(n=2)}{P(n=1)}$ starts to become significant.

13.4.1/ (5) If we neglect interelectron interaction and change the potential to $\frac{Ze}{r}$, we change $V \rightarrow \frac{Ze^2}{r}$. Thus, we should replace $e^2 \rightarrow Ze^2$ in all equations. Then

$$E_n = -\frac{me^4}{2\hbar^2} \frac{1}{n^2} \rightarrow -\frac{m(Ze^2)^2}{2\hbar^2} \frac{1}{n^2} = Z^2 \left[-\frac{me^4}{2\hbar^2} \frac{1}{n^2} \right]$$

This is because 1) the interaction is stronger: $\phi = \frac{e^2}{r} \rightarrow \frac{Ze^2}{r}$.

2) Distance is decreased: $a_0 = \frac{\hbar^2}{me^2} \rightarrow \frac{\hbar^2}{mZe^2}$

$\Rightarrow$ The total effect is E_n increases by Z^2

13.4.2/ (5) For uranium, $Z = 92$. $a_0 = \frac{a_0}{Z}$

If electrons fill in the order of increasing n , we know the degeneracy for each n is $2 \times \sum_{l=0}^{n-1} (2l+1) = 2n^2$. Then from

$$\begin{cases} \sum_{n=1}^5 2n^2 = 110, & \text{we can take the radius of uranium atom} \\ \sum_{n=1}^4 2n^2 = 60 & \end{cases}$$

to be $\langle r \rangle = n^2 \frac{a_0}{Z} = \frac{5^2}{92} a_0 = \frac{25}{92} a_0$, the

average radius of the 5th level

(5) For visible light,
 13.4.3/ $\omega_{nn'} = \frac{2\pi}{T} = \frac{2\pi c}{\lambda} \approx 3.6 \times 10^{15} \text{ s}^{-1}$

Then,

$$\omega_{nn'} = \frac{R_y}{\hbar} \left(\frac{1}{n'^2} - \frac{1}{n^2} \right) \approx 20 \times 10^{15} \text{ s}^{-1}$$

Lyman: $\left(\frac{1}{1^2} - \frac{1}{2^2} \right) = 0.75$

Balmer: $\left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5}{36} \sim \frac{1}{7}$

Paschen: $\left(\frac{1}{3^2} - \frac{1}{4^2} \right) = \frac{7}{144} \sim \frac{1}{20}$

Thus, Balmer series is most likely the first series discovered